Utrecht University

TU/e EINDHOVEN UNIVERSITY OF TECHNOLOGY

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Waterproof demo

Teaching with a proof assistant

Pim Otte
PhD Candidate

Outline

Motivation

Demo

Conclusion

Questions

What is Waterproof?

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  - Transferable to paper.

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- Examples: Talk by Terrence Tao [Tao24], AlphaProof [At24].

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- Look for checks in times of uncertainty.

References

- AlphaProof and AlphaGeometry teams, *Ai achieves silver-medal standard solving international mathematical olympiad problems*, 2024, `https://deepmind.google/discover/blog/ai-solves-imo-problems-at-silver-medal-level/`.
- Terrence Tao, *The potential for ai in science and mathematics*, 2024, `https://www.youtube.com/watch?v=_sTDSO74D8Q`.
- Jelle Wemmenhove, Dick Arends, Thijs Beurskens, Maitreyee Bhaid, Sean McCarren, Jan Moraal, Diego Rivera Garrido, David Tuin, Malcolm Vassallo, Pieter Wils, and Jim Portegies, *Waterproof: Educational software for learning how to write mathematical proofs*, Proceedings 12th International Workshop on Theorem proving components for Educational software. (ThEdu'23) (Julien Narboux, Walther Neuper, and Pedro Quaresma, eds.), 2024.

Questions?



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Sharing science,
shaping tomorrow

Joint work with Jim Portegies, Jelle Wemmenhoven and various teams of student programmers

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pim.otte.dev